

AI Reference

This file records summarized AI assistance for Assignment 02 Web Mario.

AI Tools Used

- Tool: Codex
- Usage type: planning, TypeScript implementation assistance, Cocos Editor setup guidance, documentation drafting, and commit message/description drafting
- Output format: summarized records, not full chat logs

Scope Of AI Usage

Codex was used to help with: All code

- Reading the assignment plan and local project structure.
- Planning implementation phases.
- Editing existing TypeScript files for game flow, player control, gameplay rules, animation hooks, sound hooks, and UI hooks.
- Updating `PLAN.md` after phases.
- Drafting and updating README and AI reference Markdown.
- Drafting commit messages and descriptions.

The student remains responsible for:

- Creating Cocos Editor nodes and scenes.
- Assigning Inspector references.
- Creating animation clips.
- Assigning audio clips.
- Building and deploying the web version.
- Packaging, MD5 checksum, and final submission.

File References

Game Process

- `assets/Scripts/scenes/StartMenu.ts:9`
 - Loads Level select through `goToLevelSelect()`.
- `assets/Scripts/scenes/LevelSelect.ts:9`
 - Loads Game scene through `startLevelOne()`.
- `assets/Scripts/scenes/GameOver.ts:32`
 - Restarts Game scene.
- `assets/Scripts/scenes/GameOver.ts:36`
 - Returns to Start menu.
- `assets/Scripts/scenes/StartMenu.ts:12`
 - Inspector field for the leaderboard label.
- `assets/Scripts/scenes/StartMenu.ts:40`
 - Displays the Top 5 local leaderboard.
- `assets/Scripts/scenes/GameOver.ts:19`
 - Inspector field for the final time label.
- `assets/Scripts/scenes/GameOver.ts:22`
 - Inspector field for the game-over title label.
- `assets/Scripts/scenes/GameOver.ts:25`
 - Inspector field for the name input.
- `assets/Scripts/scenes/GameOver.ts:40`
 - Public `submitScore()` method for the Submit button.
- `assets/Scripts/scenes/GameOver.ts:59`
 - Updates the title to `FINISH` or `GAME OVER`.
- `assets/Scripts/scenes/GameOver.ts:131`
 - Hides the name input after failed runs.

Player, Physics, Audio, UI

- `assets/Scripts/PlayerController.ts:27`
 - Inspector field for player animation.
- `assets/Scripts/PlayerController.ts:40`
 - Inspector field for jump sound effect.
- `assets/Scripts/PlayerController.ts:190`
 - Plays jump sound after a valid jump.
- `assets/Scripts/PlayerController.ts:225`
 - Checks whether an animation state exists before playing it.
- `assets/Scripts/PlayerController.ts:230`
 - Plays the selected player animation clip.
- `assets/Scripts/PlayerController.ts:115`
 - Pauses player motion during enemy contact.
- `assets/Scripts/PlayerController.ts:202`

- Calls `GameManager.finishLevel()` after finish trigger contact.
- `assets/Scripts/GameManager.ts:23`
 - Out-of-bounds deathY threshold.
- `assets/Scripts/GameManager.ts:32`
 - Enemy hit pause delay.
- `assets/Scripts/GameManager.ts:41`
 - BGM Inspector field.
- `assets/Scripts/GameManager.ts:44`
 - Death sound effect Inspector field.
- `assets/Scripts/GameManager.ts:53`
 - Life label Inspector field.
- `assets/Scripts/GameManager.ts:56`
 - Score label Inspector field.
- `assets/Scripts/GameManager.ts:59`
 - Timer label Inspector field.
- `assets/Scripts/GameManager.ts:98`
 - Detects out-of-bounds player fall.
- `assets/Scripts/GameManager.ts:102`
 - Public `hurtPlayer()` method for enemy damage.
- `assets/Scripts/GameManager.ts:122`
 - Public `finishLevel()` method for finish trigger completion.
- `assets/Scripts/GameManager.ts:169`
 - Saves final score, time, finished state, and result title to local storage.
- `assets/Scripts/GameManager.ts:180`
 - Pauses player motion during enemy contact.
- `assets/Scripts/GameManager.ts:191`
 - Resumes the paused enemy after player respawn.
- `assets/Scripts/GameManager.ts:219`
 - Starts the optional moving platform action.
- `assets/Scripts/GameManager.ts:112`
 - Public `addScore(points)` method.
- `assets/Scripts/GameManager.ts:237`
 - Starts looping BGM with `playMusic`.
- `assets/Scripts/GameManager.ts:245`
 - Plays one-shot sound effects with `playEffect`.
- `assets/Scripts/GameManager.ts:254`
 - Updates life UI text.
- `assets/Scripts/GameManager.ts:262`
 - Updates score UI text.
- `assets/Scripts/GameManager.ts:270`
 - Updates timer UI text.
- `assets/Scripts/CameraFollow.ts:4`
 - Camera follow component.

Gameplay Objects

- `assets/Scripts/gameplay/Enemy.ts:18`
 - Inspector field for enemy animation.
- `assets/Scripts/gameplay/Enemy.ts:13`
 - Fixed Goomba falling speed.
- `assets/Scripts/gameplay/Enemy.ts:25`
 - Goomba fly animation clip name.
- `assets/Scripts/gameplay/Enemy.ts:37`
 - Enemy score value.
- `assets/Scripts/gameplay/Enemy.ts:73`
 - Clamps Goomba falling speed.
- `assets/Scripts/gameplay/Enemy.ts:97`
 - Adds score after stomping an enemy.
- `assets/Scripts/gameplay/Enemy.ts:105`
 - Pauses Goomba before notifying `GameManager` about side contact.
- `assets/Scripts/gameplay/Enemy.ts:118`
 - Resumes Goomba movement after player respawn.
- `assets/Scripts/gameplay/Enemy.ts:153`
 - Plays the fly animation while Goomba is moving vertically.
- `assets/Scripts/gameplay/Enemy.ts:160`
 - Plays enemy animation clip.
- `assets/Scripts/gameplay/Enemy.ts:187`
 - Calls `GameManager.addScore(points)`.
- `assets/Scripts/gameplay/QuestionBlock.ts:19`
 - Question block score value.
- `assets/Scripts/gameplay/QuestionBlock.ts:41`
 - Adds score when the block is hit from below.
- `assets/Scripts/gameplay/QuestionBlock.ts:66`
 - Plays item appear sound effect.
- `assets/Scripts/gameplay/QuestionBlock.ts:81`
 - Calls `GameManager.addScore(points)`.
- `assets/Scripts/gameplay/PowerUp.ts:13`
 - Power-up score value.
- `assets/Scripts/gameplay/PowerUp.ts:51`

- Plays power-up sound effect.
- assets/Scripts/gameplay/PowerUp.ts:52
 - Adds score after collecting the power-up.
- assets/Scripts/gameplay/PowerUp.ts:78
 - Calls GameManager.addScore(points) .

Planning And Documentation

- PLAN.md
 - Records phase status, Codex-owned files, user-owned Cocos GUI setup, and implemented TypeScript behavior.
- README.md
 - Summarizes the project, controls, scoring focus, implemented features, remaining work, and AI usage.
- AI_reference.md
 - Records summarized AI usage.

Prompt And Response Records

Project Constraints

- Prompt summary:
 - The user asked Codex to act as a Cocos Creator 2.4.8 Web Mario assignment assistant.
 - The user specified that Codex should modify only existing .ts and .md files unless explicitly allowed.
 - The user also specified that Cocos Editor nodes, components, Inspector dragging, scenes, and new asset setup are user-owned.
- Response summary:
 - Codex inspected local files and planned work around existing scripts and PLAN.md .
 - Codex kept .fire , .meta , image, audio, and generated Cocos files untouched.

Phase 2 And 3

- Prompt summary:
 - Build a minimum playable version and game process.
- Response summary:
 - Codex helped prepare TypeScript for player movement, camera follow, physics manager setup, respawn, scene navigation, and game-over flow.
 - Codex listed required Cocos Inspector wiring in PLAN.md .

Phase 4

- Prompt summary:
 - Implement Basic Rules: physics, player, enemy, question block, mushroom.
- Response summary:
 - Codex helped implement enemy patrol, stomp detection, player damage, question block activation, mushroom movement, and player growth.
 - Codex updated PLAN.md with TypeScript and Editor responsibilities.

Phase 5 And 6

- Prompt summary:
 - Implement animation and sound hooks while leaving Animation Clip creation and audio assignment to the Cocos Editor.
- Response summary:
 - Codex added animation clip-name fields and safe cc.Animation playback.
 - Codex added audio clip fields for BGM, jump, death, stomp, item appear, and power-up effects.
 - Codex used playMusic for BGM and playEffect for one-shot sound effects.

Phase 7

- Prompt summary:
 - Implement UI for life, score, and timer.
- Response summary:
 - Codex added cc.Label fields to GameManager .
 - Codex added life, score, and count-up timer UI updates.
 - Codex added score hooks for enemy stomp, question block hit, and power-up collection.

Phase 9

- Prompt summary:
 - Create README and AI reference documentation in Markdown.
 - Use summarized AI records instead of full chat logs.
- Response summary:
 - Codex drafted README.md .
 - Codex drafted AI_reference.md with tool, scope, file references, prompt summaries, response summaries, and modification records.

Later Gameplay And Documentation Updates

- Prompt summary:
 - Add finish trigger behavior, final score display, leaderboard/name-input planning, moving platform, Goomba airborne behavior, and enemy-contact pause.

- Update `PLAN.md`, `README.md`, and AI reference documentation to match the latest project state.
- Response summary:
 - Codex added finish trigger hooks and last-result local storage.
 - Codex added local leaderboard hooks with name input and score/time sorting.
 - Codex added Goomba `goomba_fly` hook and fixed falling speed.
 - Codex added a short pause for Player and Goomba after side enemy contact.
 - Codex restructured `README.md` around the assignment scoring focus.
 - Codex updated `PLAN.md` to track remaining Cocos Inspector tasks and scoring progress.

Final Result UI Updates

- Prompt summary:
 - Change the game-over title after a successful finish and hide the name input after failed runs.
 - Keep project documentation synchronized with the current scripts.
- Response summary:
 - Codex updated the last-result data with a `FINISH / GAME OVER` title.
 - Codex updated `GameOver.ts` so the title label reflects the result state.
 - Codex hid the name input after failed runs and kept it visible only after finished runs.
 - Codex updated Markdown documentation and refreshed source line references.

Version And Modification Record

- 334279d Set up initial Cocos scenes
 - Initial Cocos scene setup.
- c645913 Implement scene flow controllers for phase 3
 - Scene flow controller implementation.
- a00cd40 Implement basic gameplay rules for phase 4
 - Basic rules implementation for player, enemy, question block, and power-up.
- 5fea1cc Add animation and sound hooks for phases 5 and 6
 - Animation and audio hooks.
- 2be8ff4 Add Phase 7 UI hooks for life score and timer
 - Life, score, and timer UI hooks.
- 37bae88 `fin` detect
 - Finish trigger detection and game-over transition.
- e42dee3 `game over` score, font wall
 - Final score display updates.
- 917c5e5 moving plat, up score
 - Moving platform and score-related updates.
- a9103ac Add leaderboard and name input for game over
 - Local leaderboard and player name input hooks.
- 4d3905f Add Goomba airborne animation behavior
 - Goomba fly animation and fixed falling speed behavior.
- defb660 Pause player and enemy briefly on contact
 - Short pause when Player touches an enemy from the side.
- 5ae9148 Update `PLAN.md`
 - Updated planning checklist and scoring documentation.
- 1ec804d Revise docs around scoring focus and AI reference
 - Documentation updates for scoring focus and AI reference.
- 752507c Add score and level clear sound hooks
 - Added score and level clear audio hooks.
- 9e906b5 Update HUD and game over result handling
 - Updated game-over result handling and related documentation.
- 969a733 Update deployment assets and `README`
 - Updated deployment assets and `README`.
- 65c691a `v1`
 - Final `README` update.

Notes For Final Submission

- If the TA requires `AI_reference.pdf`, export this Markdown file to PDF before submission.
- The PDF should be visually checked before packaging.
- The final package should still include source files, `README`, assets, and required build files according to the assignment rules.